

MATE 2026 REGIONAL ROV COMPETITION



TigerSharks ROV CO

Mamaroneck High School, Mamaroneck, New York, United States of America

Technical Report

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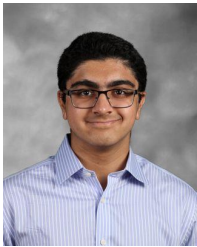


I. Abstract

We are the Mamaroneck TigerSharks ROV Company from Mamaroneck, New York. We work at Mamaroneck High School to develop our ROV to maximize task completion. These tasks model marine research, and showcase how teams would approach similar problems in the real world. For our ROV, company members worked together over the course of many weeks to make an ROV that was maneuverable and relatively inexpensive. Once we finished our design process, we methodically assembled our ROV and used rapid prototyping to iterate quickly. Our main focus this year was to expand on our previous designs and increase reliability, usability, and efficiency. We optimized electrical systems, redesigned our command and control structure for increased robustness, and added tools like a recoverable line to our ROV to increase potential workload. Our company also designed and built a non-rov device (NRD) to allow for autonomous monitoring and water state awareness.



II. Company Members

	Mustafa Khan - Senior President Mustafa is currently in his first year of computer science and played a key role in CAD and fabrication of parts. Mustafa is also part of the FTC robotics club, in which he mostly contributed to systems integration and CAD of the main robot. Outside of MATE, Mustafa is part of the MHS debate team and other computer science related initiatives.
	Max Prager - Senior Lead Software Max is in the computer science program at MHS. He worked to design some aspects of the software and hardware of the MATE robot. He enjoys programming and designing in his free time outside of school. He also participates in other stem related programs at MHS, such as FTC robotics.
	Jordan Metsch - Sophomore Lead Engineering Jordan is a Sophomore at MHS. He joined the MATE team after learning about it when competing in the FTC competition in the MHS Robotics Club. When not participating in robotics, he enjoys completing engineering projects and 3D printing.
	Ethan Bregman - Sophomore Engineering Ethan is a Sophomore at MHS, and the MATE ROV competition is his second robotics competition. Co-leading the design and production of the Float, he has learned a lot in the past few months as part of Mamaroneck Robotics. In his downtime he enjoys rocketry and building remote-controlled aircraft and running for Mamaroneck XC/TF.
	Hugh Scobie - Junior Outreach Hugh is a Junior at MHS who joined MATE after competing in the FTC Robotics competition in MHS robotics Club. He helped get sponsors for the Team. He enjoys breaking and building things and has a huge interest in Robotics and History. He also rock climbs in his free time.



	Leo Porat - Junior Engineering Leo is a junior and a member of the four-year engineering track at MHS. He joined the MATE team after learning about it from his teacher and peers. He helped primarily with the design and iteration of the claw subsystem. He enjoys mechanisms and linkages, and is interested in mechanical engineering. Outside of engineering, he also plays the drums and studies mathematics.
	Jonah Latzman - Junior Engineering Jonah is a Junior at MHS and a member of the engineering track. He joined the MATE team after learning about it from his teacher and peers. He has experience with Arduino and Python through three years of engineering coursework and contributed to the design and fabrication of robot components.
	James Abend - Junior Software James is a Junior in the computer science path in the MHS STEAM elective program. He worked in the team throughout FTC and MATE primarily through coding of the autonomous and camera systems respectively. However, he also enjoyed helping out his teammates whenever they needed an extra hand. He has an interest in most things STEAM related, particularly in coding, engineering, and medicine. He also has experience in the Mamaroneck Coding Club.
	Ari Steinberg - Freshman Software Ari is a freshman at MHS who has participated in MATE along with FTC this school year. Ari is working on learning the background of the Arduino IDE code, along with developing the ROV we are using for the competition. Outside of robotics, Ari enjoys playing the clarinet, and learning new things!
	Michal (Misha) Vaculin - Freshman Software Misha is a freshman at MHS who joined the Mamaroneck Robotics (FTC and MATE) team after realizing his love for engineering and computer science during his two years in middle school FLL robotics. This season, he focused on designing the graphical user robot-camera interface and learning computer science principles to help lead the computer science section next year. Outside of robotics, Misha enjoys piano, mathematics, and tennis.



	Brian Orozco - Freshman Engineering Brian is in his first year in high school and is in the MATE robotics club. He is a member that helps when it is needed. Outside of school he plays on a soccer team and likes to cook dinner at home. He has a lot of team spirit and is always willing to participate.
	Violet Barber-Pertuch - Freshman Engineering Violet is a freshman at MHS who is currently participating in the school's science research program and STEAM program. She participates in this year's FTC robotics competition and MATE. She contributes to each competition by using CAD to make specialized parts and 3D printing components for subsystems, particularly the recoverable line in the case of MATE. Outside robotics, she sings, plays piano & drums, and enjoys learning about quantum mechanics.
	Finn Schweter - Junior Engineering Finn is a junior at MHS who joined the MATE robotics club when he was a freshman. This season Finn contributed to the designing of the ROV, specifically using Fusion 360, to 3D print parts. Outside of robotics, Finn plays tennis and jazz trumpet.
	David Torres-Agluiere - Freshman Outreach David is a freshman at MHS who joined MATE in the MHS robotics club. He contributed to the final design of the ROV. He enjoys playing the trumpet, fixing computers in the computer repair club, and competing on the varsity fencing team.
	James Love - Engineering Teacher Mentor Mr. Love has degrees in Environmental Engineering and Education. When not teaching, he enjoys breaking and building things. He also enjoys snowboarding, hiking, and sailing.



III. Missions

Seabed 2030: A Kaleidoscope of Corals in Cold Water

The company must protect the deep sea ecosystem, by solving real world problems that exist in the deep ocean. This ecosystem contains four challenges our company has to overcome; starting with the savior of coral and stars in the sea. After identifying organisms, the company must traverse the ocean, to map the ocean floor, in order to figure out the sizing of the coral garden. The company then must. Once we map the coral garden, the company must take a video of the garden, and check for danger. This allowed our company to save the coral, and allow their home to be safe from predators, and the horrors of global warming.

SmartAtlantic Alliance: Better Information, Better Decisions

The company begins by discovering how many European Green crabs are located in The ocean. After figuring out the amount of invasive species, the company would track the iceberg location, which would begin to affect the oil platforms. The company would then measure the keel depth of the iceberg, and figure out if the threat of the iceberg would impact the oil platforms. Next, the company would test if the fishing gear they used was effective in terms of whale safety, where they would recover the lobster pot. Once the company tested whale safety, they would recover the buoy anchorer, retrieve the eDNA sensor, and install a new one.

Wind-Powered Offshore Oil Platform: Scalable Solutions for Global Energy Needs

This task begins by helping the ecosystem, by installing a micropile in order to support the plant growth. The company needs to place a device around the location for the micropile, guide the micropile, and then finally place it in the desired location. Nearby, an oil platform needs a source of energy. To solve this issue, a wind turbine is required to be installed. First, a power connector is retrieved from the wind farm subsea station, and the power connector is then navigated away from the micropile, in order to place the connector into the platform's port.

MATE Floats!

The company must construct a vertical profiling float, which communicates with the mission station prior to descending. Then, the float must complete two vertical profiles, with or without a buoyancy engine, with each repetition communicating the time, and temperature to the mission station. The company has to travel from the surface of the pool, to the top, and back, which is all documented with the use, safety, and goals of the float.

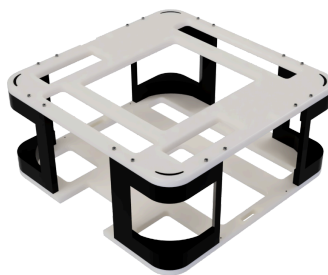


IV. Teamwork

Being a school team, the time that our company had to work on our ROV is a precious and limited resource. At the beginning of our season, our company created a Linear account, in order to help plan out deadlines, manage our schedule, and effectively communicate with each other. This app featured everything from when individual designs were due to be completed, to the date we ordered specific parts. Utilizing this app, the company was able to efficiently manage our time and delegate tasks to specific members. Our company also utilized many online tools to efficiently communicate and collaborate. These tools include the previously mentioned Linear as well as the companies' Google Drive to write our technical documents, specification sheet, and to design our posters collaboratively. For design and development, we use GitHub and Fusion360 extensively to ensure that all of our team members stay up to date with technical specifications and developmental changes. As with any project where multiple people are working together on one final product, our company had many differing views on topics as large as the overall design of the ROV to the smallest details. While sometimes these differing views lead us to creating unique solutions to problems that arise throughout the season, they can also lead to problems with integrating parts of the ROV together as a final product. Our company overcame this problem by keeping in constant communication with each other as problems arise, and by making sure that our engineers have relatively similar views on how we design individual components and how they link together.

V. Design Rationale

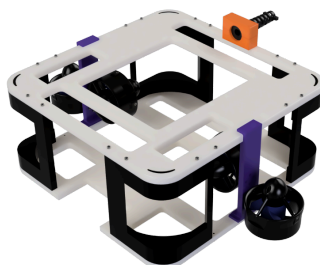
Structure



The main structure of the frame consists of Delrin and Tough PLA. These materials were selected after extensive research based on density, weight, machinability, and price. The top and bottom plates went through many iterations and were cut out on a CNC router. The corner braces were 3D printed out of Tough PLA and attached to the Delrin using heat set threaded inserts and M3 bolts. The corner braces were designed to reduce turbulent flow in the water and minimize mass while ensuring structural stability. We designed our frame to balance structural integrity with generalizability (numerous screw holes and gaps) to enable easy re-usage throughout different seasons by simply adding different modularizable attachments. This allowed us to save major time and money in maintaining a major part of our ROV consistent throughout the years.



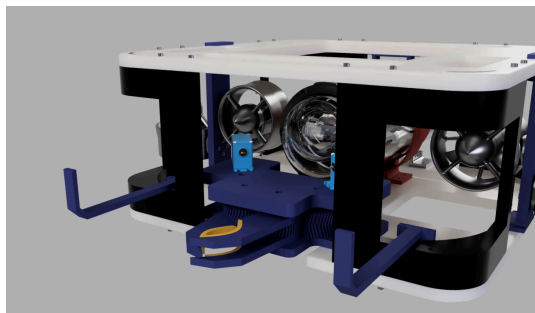
Propulsion and Buoyancy



The thruster layout was a big factor during the initial design stage. At first, engineers focused on mounting four thrusters in the corners, all rotated 45° from the center lines, to allow for omnidirectional movement. Later in the design process, the company settled on a more stable layout, having two facing up and two facing the front; this layout was chosen for its flexibility, balance, and ease of control. The layout is such that the downward vector of the upward-facing thruster cannot interfere with the front-facing thruster, an important factor for ensuring reliable movement. Also, with the upward thruster locations being within the frame and pointing towards cutouts in the delrin, more efficient use of space and allows the center of thrust to be as close to the center of mass as possible. The thruster of choice for our company was the Blue Robotics T200, which was picked for its high performance and cost effectiveness over time. In terms of performance, the T200 has a high thrust-weight ratio and high efficiency of over 23.3 grams/Watts – better than many high-end industrial thrusters (around 15-20 grams/Watts). Cost wise, despite its relatively high initial cost (~\$270), its superior longevity (600–1000+ hours) makes it far cheaper over many seasons. To comply with safety regulations, we soft-locked the motors into a thrust range where they can only operate with a maximum of 8 amps. When designing for buoyancy, the company engineers decided on an ROV that is slightly positively buoyant. This would act as a safety measure and ensure the ROV would float to the surface in the case of any onboard failure.



Claw

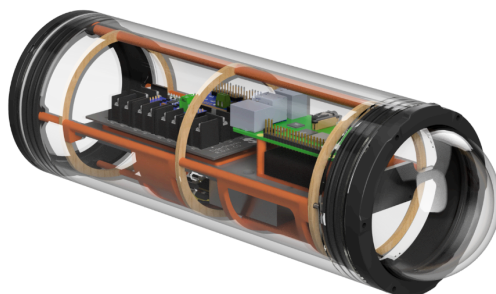


The design of the claw stayed mostly consistent with the previous season due to the adaptability of the claw. This season, as many missions require a claw, the company's focus was to make the claw as robust as possible. The claw is 3D printed with PLA filament, due to its rigidity, precision, and ease of printing. Double helical gears were chosen to reduce backlash, prevent gear slippage, increase the strength of the gear system without introducing uneven vertical forces (such as standard helical gears). The most important feature that ensures the claw is as strong as possible is the placement of the servos on auxiliary gears, away from the main gears of the claw gripper. Not only do these gears allow the claw to have more torque, but they also shield the servos from radial or axial loads, ensuring that the spline or shaft of the servos is not damaged. The gears attached to the gripper and their housing absorb the force from any accidents, impacts, or motion, rather than the spline of the servo. The company also carefully optimized the design of the claw jaws to be as effective as possible. The claw has been designed to grab and pick up PVC piping as well as loops of wire or plastic attached to other objects. For this reason, an interlocking claw design was chosen, where the jaws of the claw overlap in the horizontal plane, so that no small objects can fall out of the claw's grip. Towards the center of the claw, the claw grip widens and curves, such that a variety of tubes of different diameters can be effectively grasped. For additional grip strength, a compressible interface was added between the curved section of the claw and objects to be picked up, made of flexible TPU filament with gyroid infill for maximum compression. To ensure that the jaws of the claw always stayed aligned and the servos could share any torque applied, the two grippers on the claw are geared to each other as well as their respective servo. This distributes torque from the servos and prevents any issues with misalignment during the product demonstration. Finally, to maximize the torque of the claw, two servos were used, alongside a 2:1 gear ratio between the servos and the gripper.

This year, we modified our claw slightly by adjusting the grippable diameter from 2in to 2.3in. Although a 2 inch claw was highly suitable for last year's missions (thin T-handle hooks, rope loops, and small-diameter PVC pipes), this year's missions rely heavily on thicker PVC pipe bodies. We analyzed the trade offs between costs to modify the claw and the amounts of missions we could accomplish, and we realized that if we didn't switch out the claw, we would have been physically incapable or severely hindered in completing missions Task 1: Coral Garden Species Collection, and Task 2: Servicing the Holyrood Subsea Observatory. Therefore, we decided to modify the claw, and to minimize the costs, we kept the hook body and inner 3D-printed gears, and only reprinted the modularizable claw.



Electronics



The ROV contains an acrylic enclosure that holds the Raspberry Pi, 2 cameras, ESCs (Electronic Speed Controllers), buck converters, and an Arduino. The wires from the thrusters are fed through water tight penetrators and wired into the ESCs, which output the proper 3-phase power required. The ESCs use 12v power and PWM (Pulse-Width Modulation) signals to control the speed of the thrusters. The Arduino communicates with the PWM board through I2C which outputs PWM signals to the ESC's and allows for rapid movement of the thrusters. The PWM board allows us to have more outputs than are traditionally available on an Arduino. Both the Arduino and RPI use 5v power which is supplied by a buck converter positioned below the RPI. The Thrusters get power over a parallel conductor input, split over multiple power lines joined at the ends, in order to limit voltage drop over the length of the tether.

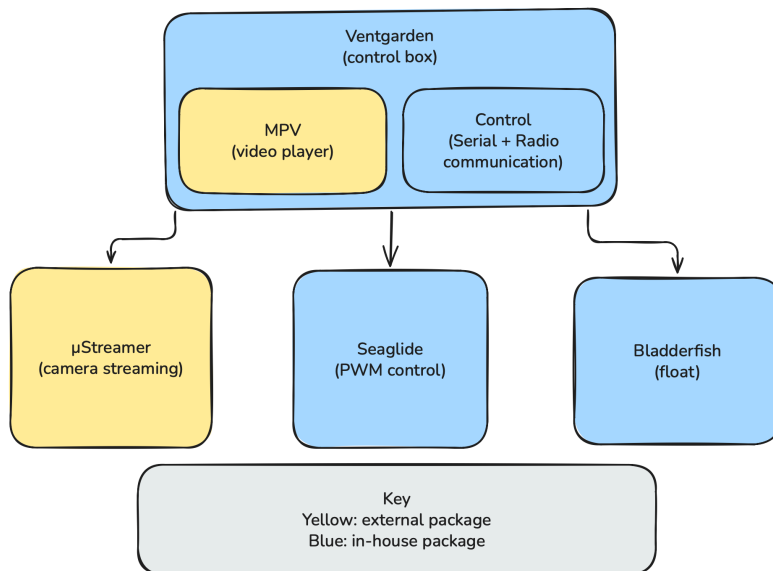
A large part of the electronic container's design was that we needed to fit the Pi, cameras, ESCs, buck converters, and Arduino. The container not only needed to be compact, but it needed to be organized, in order to avoid having shorted electronics. To solve this issue, Fusion360 was utilized, as a virtual organization tool, which provided the best way to fit each piece.

The entire tether is similar to what we used last year and consists of 5 total wires, an ethernet cable for sending video from the RPI, three total power wires to provide power to the buck converter and thrusters, and an RS-485 cable which provides controller data to the arduino. The reason we kept the tether mostly without modification is because it successfully accomplished its goals, had strong data integrity, and had little voltage drop. The ROV features an onboard strain relief hub to prevent damage to the cables and structure during operation. As an additional measure to guarantee stable and repeatable movements we evenly secured foam across the length of the tether in order to ensure it is naturally buoyant. This year, we chose to wrap our tether in a mesh split loom in order to make it more organized and easier to carry.

Finally, there are two 3D-Printed camera mounts. The primary camera is a Logitech C920, chosen for its broad field of view, high resolution, and good lowlight performance, and is located at the dome in the front of the electronics enclosure and allows us to look straight on. There is an additional camera in the middle of the enclosure positioned to point downwards, which allows us to accomplish the flythrough mission. We chose the Logitech C270 for its more focused field of view and its compact size allowing for greatest proximity to the bottom of the enclosure tube (and the bottom of the pool).



Control System



Based on the discussed design choices for the physical design of our ROV, the control system had to accomplish three primary tasks: send camera data, to be able to see what the ROV is doing, send PWM signals to the required hardware devices (servos or thruster ESCs), and transmit radio signals. A digital control system was chosen for a few reasons. Primarily, it makes our control system modular and extensible, as we are able to rapidly iterate and add features. Similarly, with many tasks requiring data analysis, processing, or storage, a digital control system allows our company to build up from our base software control system to easily add such features. The rationale for our use of a digital control system is similar but stronger compared to our last ROV, as the current set of tasks requires even more digital analysis (flyover recording, still capture, photogrammetry, and automatic counting, etc).

For ROV control, we separate the tasks of camera transmission and PWM (motor / servo) control. This allows us more flexibility to choose the individually best solution for each task, while also increasing reliability by reducing interdependence. We use a Raspberry Pi 4 on the ROV as the core of our control station, and another Pi 4 on the ROV. The Pi on the ROV solely transmits camera data using the external μStreamer package via the Ethernet. This package transmits MJPEG camera data, which substantially reduces our latency, by using hardware JPEG compression on the camera. This technique, combined with using MPV and OpenGL hardware accelerated rendering in Ventgarden (control station) allowed us to substantially reduce our camera latency, from ~280ms on our old, entirely custom control system, to now ~90ms. We can also run both the upper and lower camera simultaneously at 1080p 30fps, as separate, parallel, μStreamer processes are used for each camera. However, we limit the resolution to 720p, as there isn't any noticeable improvement in controllability with 1080p over 720p.



For PWM control, we use an Arduino and PWM driver board on the ROV to send PWM signals. To communicate with the ROV Arduino, we use the RS-485 board / protocol, for low latency, strong reliability, and good throughput for serial data communication over long distances. The RS-485 protocol is designed to operate reliably for distances over a kilometer, which makes it overkill but effective for our 50ft tether. On the control station side, Ventgarden uses a Python script to send controller data (we use a Logitech Game Controller to pilot the ROV), which goes to a secondary Arduino inside of the control box, and finally to the RS-485 cable on the tether. The same control software uses the same secondary Arduino to send and receive radio signals from the Non ROV device (the float).

The two halves of Ventgarden (MPV / video streaming, and control) run on separate processes on the Raspberry Pi, which ensures that any issues, throttling, or inconsistencies, in one system do not impact the other. One of the largest strengths of our control system is its ease of development / maintenance, and its reliability. These are enabled by the use of external software packages, which not only make our control system highly performant, but reduce the amount of code that has to be maintained internally. This meant that Ventgarden could be written in Python, which is easy to read, understand and maintain, because the Python code only controls external libraries / processes, all of which are written in C / C++ and are pre-compiled binaries. However, this use of external libraries is also one of our larger limitations - we have less flexibility in some ways - for example, we are limited in the ROV side manipulation of the camera stream that is transmitted to the topside control station. Secondly, the use of a digital control system introduces more difficulty in initial development. These drawbacks, however, were vastly outweighed by the benefits of this new control system, especially when compared to last year's. Improvements in software performance, such as a 70% improvement on camera latency to the ability to display two live 1080p and 30fps video feeds without breaking 10% CPU memory usage, confirm that our control system has undergone significant improvement. Most importantly, the simplicity of this software structure has allowed engineers on the team to allocate time to other components of the ROV, instead of spending hours on several custom built libraries using 3 separate coding languages. Reflecting on this year's Control system, our biggest improvement was simplicity.



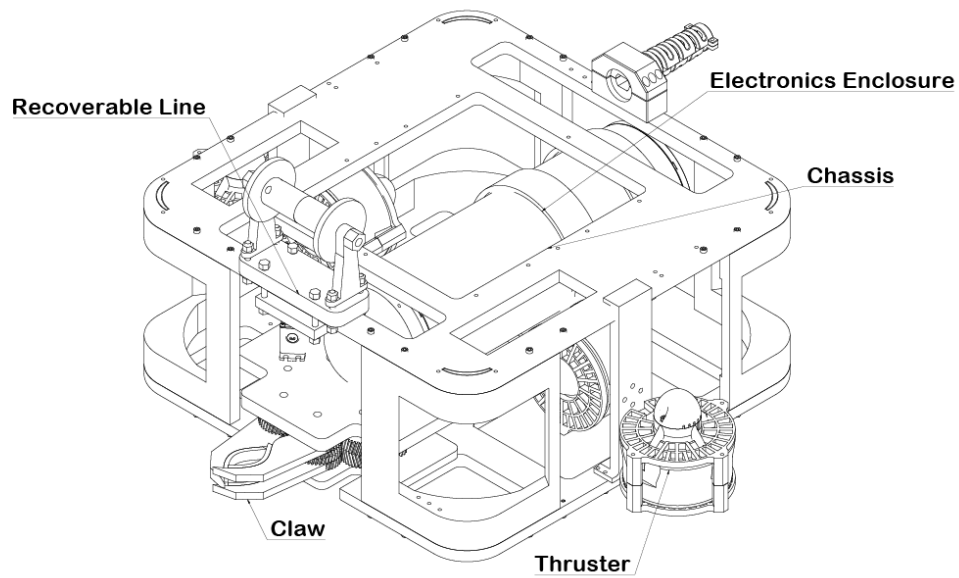
Recoverable Line



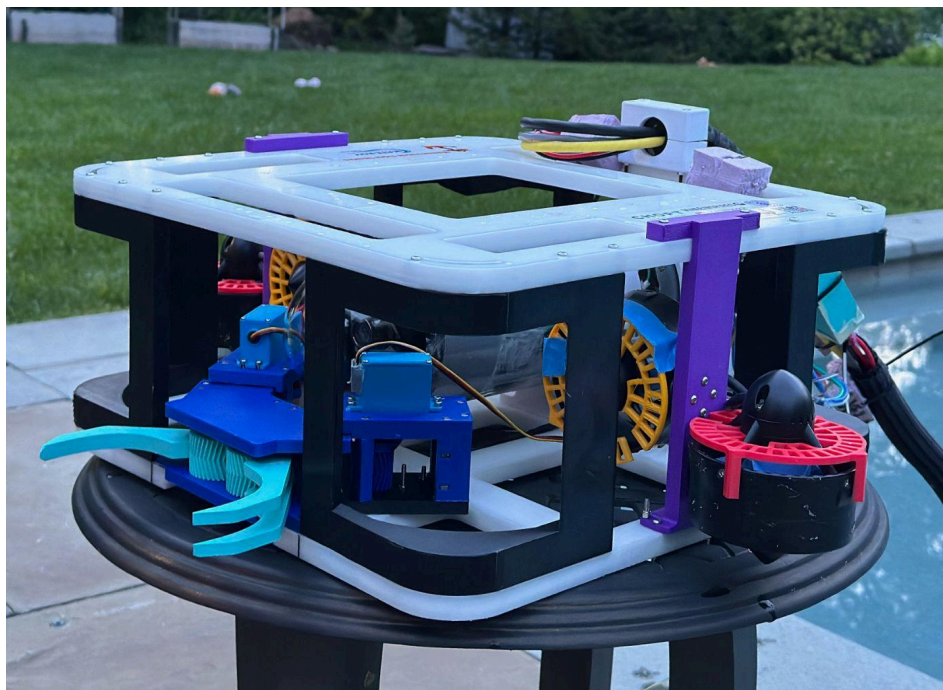
One of our main goals designing this subsystem was to keep it low in cost and versatile, while also maintaining a standard of quality that would allow us to consistently complete the task given with exceeding levels of precision. The Mamaroneck robotics team had a very logical design process for this subsystem involving sequential thought of each factor impacting the subsystem, external research, building, testing, and optimization. To start, we asked ourselves three essential questions—what do we need this subsystem to do, how can we do that, and how can we optimize the practicality of this subsystem? After brainstorming and researching ideas for how we could cost-effectively, accurately, design this subsystem we settled on making at least $\frac{2}{3}$ of our necessary components 3D printed. We decided that the remaining parts were going to be metal and nylon rope for the sake of strength and material durability. After initially assembling the parts and testing, we noticed that everything we designed worked effectively, but we as a team thought there was still room for improvement in terms of versatility. To solve this, we decided to use a clamp-like design on the bottom of the frame of the subsystem. That final addition completed our subsystem, allowing it to be easily mounted to any part of the robot.



Technical Drawing



Final ROV





VI. Accounting/Materials

Before starting the design and fabrication process, company members examined the ROV from the past year and recycled all components that were salvageable, such as the tethers and ESCs, thrusters, and the frame.

Material Expenses:

Item	Qty	Cost/Unit	Total Cost
Pressure Sensor	1	\$29.95	\$29.95
Tubing	1	\$4.49	\$4.49
Fuse	1	\$8.99	\$8.99
Camera	1	\$43.59	\$43.59
Motor Cables	1	\$7.99	\$7.99
SMA Cable	1	\$6.99	\$6.99
Lora Breakout	2	\$24.33	\$48.66
SMA for LORA	1	\$10.99	\$10.99
Tubing	1	\$4.49	\$4.49
Battery	1	\$19.99	\$19.99
Motor Driver	1	\$7.99	\$7.99
Hose clamp	1	\$5.99	\$5.99
Power Cable	1	\$17.99	\$17.99
Power Plug	1	\$9.40	\$9.40
RS485 Cable	1	\$34.65	\$34.65
RS485 Boards	1	\$8.99	\$8.99
14 Hole End Cap	1	\$16.00	\$16.00
ESC	1	\$38.00	\$38.00
Bullet Connectors	1	\$21.99	\$21.99
Wetlink Penetrator Seals	1	\$18.00	\$18.00
Blank Penetrators	3	\$6.00	\$18.00
Epoxy Penetrators	9	\$6.00	\$54.00
T200's	4	\$258.00	\$1032.00



Item	Qty	Cost/Unit	Total Cost
Keyboard/Trackpad	1	27.99	27.99
Second Power Cord	1	69.99	69.99
Watertight enclosure	1	4.99	4.99
T200 C	1	6.99	6.99
T200 CW	1	8.99	8.99
Buck Converter	1	7.99	7.99
Screw terminal	2	15.99	31.98
Resin	1	6.99	6.99
Red/Black Wires (25')	1	12.96	12.96
Control Box Plugs	2	9.20	18.40
Camera	2	29.99	29.99
Radio module	1	8.49	8.49
Delrin	1	403.58	403.58
		Sub Total:	2,108.46

Other Expenses:

Item	Total Cost
Bus	<i>Covered by School</i>
T-shirts	565.29
Competition Fee	500.00
Sponsor	300
	Sub Total:
Total Expenses	1365.29
	2873.75



VII. Constraints

When constructing the ROV we ran into several constraints. Two of the main ones being our lack of time and lack of space. Since we couldn't find a collective space to meet, we had to meet on and off at each other's houses, assembling the ROV and finding time to do all of this in between our busy individual schedules. To compensate for the little amount of time that we had we created a Microsoft Teams timeline with when each step of the process should be completed and how to adjust the schedule if something didn't go exactly to plan. A challenge that has repeatedly posed a problem throughout the past few years of our team has been that, our company systems design rationale has often led to instances where we went over our allowed maximum amperage pull. We have fixed these problems by implementing more efficient computing systems, and limiting our electrical pull through software solutions. Another primary challenge that we ran into making reliable electrical and communication systems. We solved this issue by adding 3 sets of new cables, two of which are power conductors acting as parallel conductors to reduce voltage drop. The third cable is an additional communication cable that uses the RS-485 protocol to communicate between devices reliably over long distances. Additionally, we ran into many issues for our 3D printed parts such as the pump and the claw when it comes to the durability of these parts. Finally, we ran into many problems when creating the Float, such as how to make a buoyancy engine, as this was a new challenge that we had not tried since the mission was first released a couple of years ago. The buoyancy engine promoted problems due to the company needing to figure out how to make a buoyancy that was both strong enough and waterproof. This problem was ultimately solved through the use of separate chambers, a wet chamber and a dry chamber, and also using a rotary seal to connect the two.

VIII. Critical Analysis

As with any ROV or product, issues are bound to arise during the development process, but they can be minimized. One way we decreased the likelihood of the problems is through our prototyping methods. Many of the parts are 3D printed, specifically those that connect externally to the frame. The most significant advantage of 3D printing is its speed. All of the 3D-printed components on the ROV went through multiple iterations, testing different parameters until maximum efficiency was achieved. For testing the ROV, we used a multi-stage process. Testing the buoyancy first allowed us to gradually modify the buoyancy of the ROV using foam to achieve neutrality. Later in the testing process, the ROV was placed in a pool with all its electronics, allowing our engineers to test all the thrusters. The new software control systems work in harmony to minimize the effects of errors and unexpected fluctuations. Our system design allows each system to specialize in one aspect of the ROV's task, increasing reliability and robustness. Through careful procedures, design, and implementation, the company was able to minimize errors.



IX. Challenges

Institutional Knowledge:

Institutional knowledge is something that we constantly struggle with. While our school technically has had a MATE company in the past, it was canceled before Covid-19, and as such, no one on our team had ever participated in MATE before our founding in 2022. As such, almost all of our systems have been built from the ground up using knowledge that we had from before MATE, along with knowledge that we've gained in the last two years. While admittedly, we have learned a lot in the last two years about how to manage and run a MATE team, as well as build ROV's, we still struggle with issues relating to a lack of knowledge relatively often. One such example is the Software suite we use, as outlined above, which has been made by scratch by our company members and is the culmination of everything we have learned in the last few years.

Future recruitment and potential loss of knowledge:

We anticipate difficulties in the future with our knowledge base. As shown in the members section, most of our company members are currently seniors in high school, and as such will not be here next year. We expect that this could follow with many of our non-senior members being left without knowledge bases for how to work with the software, design the robot, and test the systems. To hopefully mitigate these problems, we have begun working on comprehensive documentation of software systems, as well as encouraging non-senior members to ask as many questions as possible. Our company leadership fully expects this loss of knowledge to be a problem in the future but we are trying our best to secure against it, and set up pathways in the future to minimize the effects of these problems.



X. Safety

Safety Features:

The safety features utilized by our ROV include netting draped over the thrusters, fuses, an emergency stop, and strain relief on the tether. The 3D printed guards prevent injuries to people and damage to property by creating a barrier between the thruster and its surroundings. The fuses prevent damage to the ROV by stopping the flow of current when it exceeds the amount of current that our ROV can draw at maximum (calculations below), indicating a short. The emergency stop allows us to stop all of the ROV's systems if control over the ROV is lost by cutting power. The strain relief prevents damage to the electrical connections by securing the tether to a point on the frame, such that any strain or tension on the cable is applied to the frame rather than delicate connections.

Safety Checklist

Pre-round checklist:

- ☐ Ensure tether safety
 - ☐ Ensure all connectors are secure and tightened
 - ☐ Ensure that tether does not have knots
 - ☐ Ensure that tether is neatly coiled and easily extended/retracted
 - ☐ Ensure that all tether bindings and floats are secure
 - ☐ Ensure that Ethernet, Power, and RS485 cables has no abrasions, breaks, or other continuity errors
- ☐ Ensure team safety
 - ☐ Ensure that all team members on the pool deck are wearing appropriate attire
 - ☐ Ensure that all team members on the pool deck are wearing safety glasses
 - ☐ Ensure that all team members on the pool deck are aware of their surroundings
- ☐ Ensure ROV safety
 - ☐ Ensure that the power connector is connected securely to the battery or ground station
 - ☐ Ensure that fuse is installed, within 30cm of the ground station, and operational
 - ☐ Ensure that power switch works and is operational in emergency situations
- ☐ Initiate ROV/ground station software
 - ☐ Ensure that all control software is running nominally



- ☐ Ensure that the ground station and the ROV have a stable connection through the ground station software
- ☐ Ensure that controllers are in X-mode
- ☐ Ensure that the cameras are receiving a live video feed

During round checklist:

Software failure on ROV or ground station

- ☐ Ensure that there is a software failure and not a momentary loss of connection
 - ☐ Force stop ground station software
 - ☐ Restart ground station and ROV software
 - ☐ Make sure that company engineers are aware of the error or problem so that further diagnosis can be applied after the round

Failure of ground station or ROV hardware

- ☐ Hit the E-Stop immediately after crash
- ☐ Restart the ROV or ground station hardware
- ☐ Make sure that company engineers are aware of the error or problem so that further diagnosis can be applied after the round

Loss of control of the robot

- ☐ Hit the E-Stop **immediately**
- ☐ Check system logs for the ROV and ground stations to try to determine the exact cause of loss of control
- ☐ Follow the steps outlined in **Failure of ground station or ROV hardware or Software failure on ROV or ground station** to reset and fix what was diagnosed in the previous steps

Loss of video feed

- ☐ Put down controller and other movement sensitive equipment **immediately**
- ☐ Check system logs for the ROV and ground stations to try to determine the exact cause of loss of video feed
- ☐ Follow the steps outlined in **Failure of ground station or ROV hardware or Software failure on ROV or ground station** to reset and fix what was diagnosed in the previous steps
- ☐ Make sure that appropriate company engineers are aware of the error or problem so that further diagnosis can be applied after the round

**Post round checklist:**

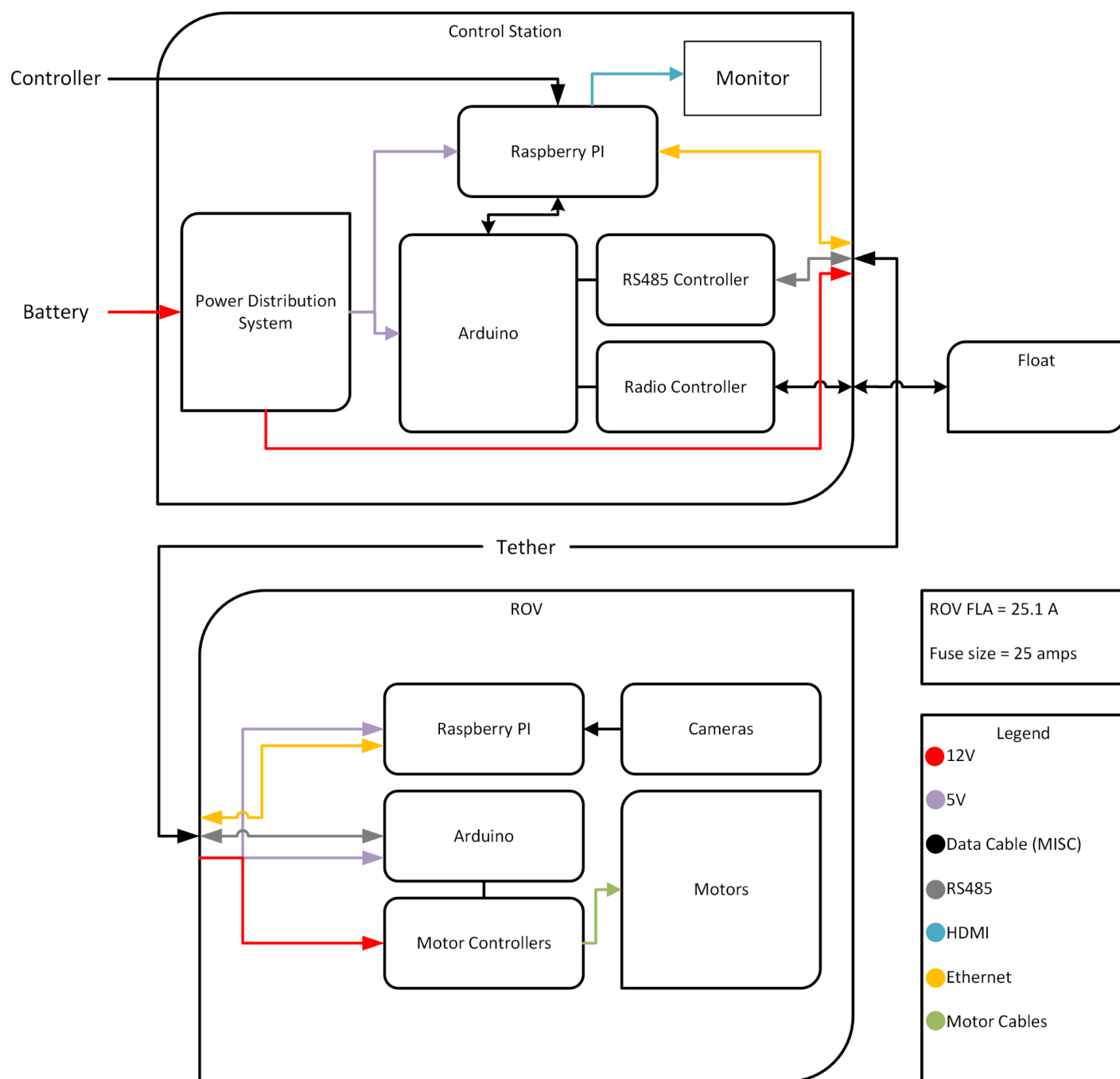
- ☐ Unscrew pressure relief valve to let latent air escape or enter
- ☐ Carefully remove control hardware from watertight tube
- ☐ Inspect for water intrusion
- ☐ Run a team meeting to diagnose and address problems encountered during round to ensure further safety during following rounds and events

Jobsite Safety and Environmental Analysis (JSEA):

We have developed a Jobsite Safety and Environmental Analysis outlining potential hazards during all times of interaction with company systems. It outlines safety measures for Testing, Bot Operations, Assembly, and Design/Development. Our goal within the JSEA was not only to ensure that all company members were operating within safe physical environments, but also that all personnel remained alert, free of fatigue, and not burnt out. It also establishes a flow and delegation of control, assigning and ensuring responsibilities with all company members. It was used during all times outlined on the JSEA, and a copy is included with this technical document.



XI. Systems Integration Diagram (SID)





XII. Sponsors & Acknowledgements

We would like to thank our sponsors, teachers, parents, and everybody else that supported us and helped our company to succeed.

Sponsors

Total Received: \$1200

	Foley's Hardware
	Decicco's and Sons
	WSP
	McMicheal's Yacht Yards and Brokers
	Makeinspires

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- Mr. Love – We would like to thank our dedicated mentor Mr. Love for his help getting us access to the school fabrication machines, helping recruiting new members, and his unwavering support for our company.

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